

UE19CS251

DESIGN AND ANALYSIS OF ALGORITHMS
UNIT 5: Limitations of Algorithmic Power and
Coping with the Limitations

All Pairs Shortest Path (Floyd's Algorithm)

PES University

Outline

Concepts covered

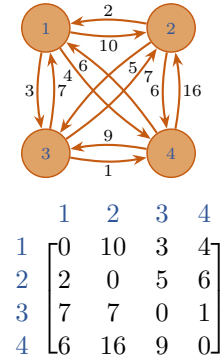
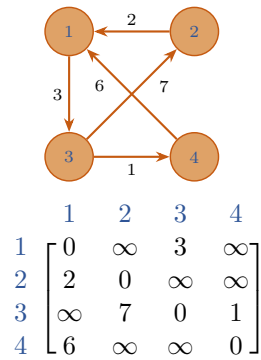
- All Pairs Shortest Path (Floyd's Algorithm)
 - Definition
 - Algorithm
 - Example

1 Problem Definition

- Given an undirected or directed graph, with weighted edges, find the shortest path between every pair of vertices
 - Dijkstra's algorithm found shortest paths from given vertex to remaining $n - 1$ vertices ($\Theta(n)$ paths)
 - Current problem is to find the shortest path between every pair of vertices ($\Theta(n^2)$ paths)
- Solution approach is similar to the transitive closure approach: Compute transitive closure via sequence of $n \times n$ matrices $R^{(0)}, \dots, R^{(k)}, \dots, R^{(n)}$ where $R^{(k)}[i, j] = 1$ iff there is nontrivial path from i to j with only first k vertices allowed as intermediate vertices
- Compute all pairs shortest paths via sequence of $n \times n$ matrices $D^{(0)}, \dots, D^{(k)}, \dots, D^{(n)}$ where $D^{(k)}[i, j]$ is the shortest path from i to j with only first k vertices allowed as intermediate vertices

2 Example

- Example of all pairs shortest paths:



3 Algorithm

Transitive Closure (Floyd's Algorithm)

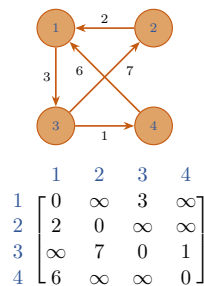
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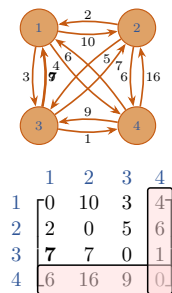
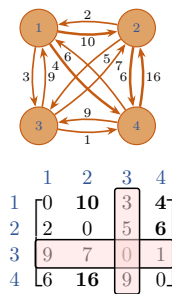
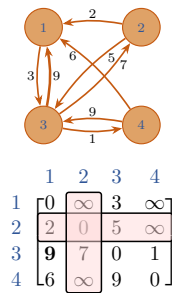
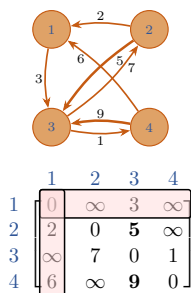
1: procedure FLOYD( $(A[1 \dots n, 1 \dots n])$ )
2:    $\triangleright$  Input: Weight matrix  $A$  of a graph with no negative length cycles
3:    $\triangleright$  Output: Distance matrix of shortest paths
4:    $D \leftarrow A$ 
5:   for  $k \leftarrow 1$  to  $n$  do
6:     for  $i \leftarrow 1$  to  $n$  do
7:       for  $j \leftarrow 1$  to  $n$  do
8:          $D[i, j] \leftarrow \min(D[i, j], D[i, k] + D[k, j])$ 
9:   return  $D$ 

```

- Complexity: $\Theta(n^3)$

4 Example





5 Think About It

- Give an example of a graph with negative weights for which Floyd's algorithm does not yield the correct result
- Enhance Floyd's algorithm so that shortest paths themselves, not just their lengths, can be found